Spatial User Interfaces: A Planet before our Time

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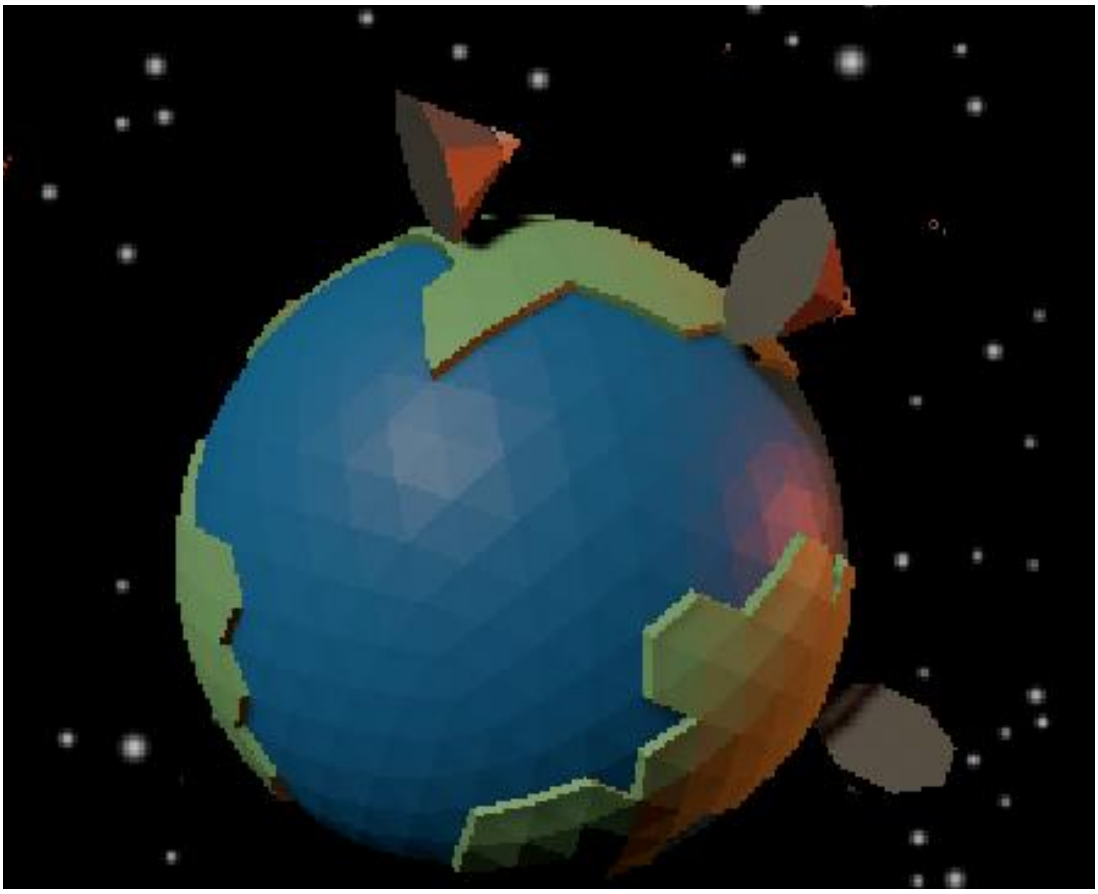

Figure 1: In-game screenshot of *A Planet before our Time*

Abstract

Pointing and grabbing are fundamental interaction mechanics in Virtual Reality (VR) experiences. With the emergence of real-time hand tracking, these interactions are evolving beyond traditional controller-based input, opening up new possibilities for a more natural and immersive user experience. *A Planet before our Time* explores this development by implementing hand tracking within a VR planet-building game, allowing users to interact with the environment using their own hands.

This paper evaluates the overall usability of the experience by analysing user feedback, providing insight into the potential controller-free VR systems.

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1 Introduction

The rise of Virtual Reality (VR) technology is creating new opportunities for more immersive digital experiences, especially through haptic mechanics. As VR systems improve, they can simulate not just sight and sound but also touch, allowing users to feel more immersed in the virtual environments.

Pointing and grabbing objects are among the most common interactions in VR, making haptic technologies essential for enabling natural user experiences. Through the sense of touch and movement, haptics help users perceive and interact with virtual environments more intuitively (S.Biswas, Y.Visell, 2021).

Typically, VR systems rely on handheld controllers that users grip to interact with the virtual world. However, in A Planet before our Time, no controllers are used. Interaction is instead performed entirely through real-time hands tracking.

This approach allows players to use their own hands directly, creating a more natural and immersive experience compared to traditional input methods (Masurovsky et al., 2020).

In addition to hand tracking, a VR headset enhances immersion by blocking out the physical surroundings and adjusts dynamically to the users movement. This combination helps create a strong sense of presence, making users feel as though they are truly part of the virtual environment (Masurovsky et al., 2020).

This project presents A Planet before our Time, a VR planet-building experience with a dinosaur theme, that relies entirely on hand tracking, without the use of traditional controllers. The interaction design focuses specifically on pointing ang grabbing mechanics, allowing users to manipulate the environment using natural hand movements. In addition, this project investigates the usability of the game by analysing user feedback and evaluating how effectively players can interact with the virtual environment.

2 Related Work

Real-time hand tracking has become an increasingly important feature in Virtual Reality (VR), playing a key role in enhancing user immersion. By allowing users to interact with virtual environments using their own hands rather than external controllers, this technology creates a more direct and natural connection between the user and the digital world. However, despite its potential, the rapid development of hand tracking also introduces a range of challenges, many of which are closely related to human factors and the complexity of accurately interpreting hand movements (Buckingham G., 2021).

The human hand is highly dexterous and capable of a wide variety of precise and subtle movements. Replicating this level of complexity in a virtual environment presents significant technical difficulties, as computer systems must track, interpret, and respond to these movements in real time. As a result, early hand tracking interfaces have often struggled with accuracy and reliability. Nevertheless, ongoing advancements in tracking technology and software algorithms are gradually reducing these limitations, enabling more precise input recognition and smoother interaction. Over time, these improvements are expected to significantly enhance the realism and responsiveness of VR experiences (Sharp et al., 2015).

Despite the technical challenges, studies have shown that the use of real-time hand tracking can positively impact the user experience. Participants have reported higher levels of emotional engagement and a greater sense of realism when interacting with virtual objects using their hands. In particular, actions such as grabbing and manipulating objects feel more natural compared to controller-based interaction. This contributes to a stronger sense of presence, where users feel more immersed and connected to the virtual environment (Voigt-Antons et al., 2020).

3 Design and Implementation

For the hardware setup, a Meta Quest 2 was used to run and support the game, providing both the display and tracking capabilities needed for an immersive VR experience. On the software side, development was carried out in Unity using Universal Render Pipeline (URP) 3D template.

To enable the core VR functionality, the XR package was integrated to handle essential features such as camera setup and spatial tracking. In addition, the Meta XR SDK was implemented to manage user interactions. This SDK supports mechanics like grabbing, selecting and gesture recognition, allowing the experience to rely on hand tracking instead of traditional controllers.

For the visual assets, models were created using Blender and then imported into Unity as prefabs. These assets were designed in a lowpoly style, to reduce computational loas and ensure smoother performance within the VR environment. This approach balances visual clarity with efficiency, making the experience more stable and accessible on the chosen hardware.

4 Evaluation

Participants were asked to evaluate the usability of the VR experience across five key categories: hand tracking, efficiency, comfort, immersion and ease of use. Each category was presented as a series of statements and participants indicated their level of agreement using a five-point Likert scale. In this scale, a rating of 1 represented strong disagreement, while a rating of 5 indicated strong agreement with the given statements.

The study focused on capturing participants' perceptions of how effectively the system supported natural interaction through hand tracking, as well as how comfortable and intuitive the experience felt during use.

A total of six participants were involved in the usability testing, all of whom were between the ages of 22 and 28. This relatively small sample size was suitable for an initial exploratory study, providing focused insights into user experience while still allowing for identifiable patterns and feedback across participants.

5 Results

The average score of 3.45 indicates a moderately positive user experience, with responses leaning slightly above neutral, suggesting acceptable usability but with room for further improvement. In addition to this, the data collected from the Likert scale questionnaire was compiled and visualized through graphs, allowing for a clearer and more accessible representation of participants responses across different usability categories.

From the results, the highest-ranking category was comfort. Achieving an average score of 4.0. This suggest that participants found this aspect of the experience to be the most successful and well-implemented. In contrast, the lowest-ranking category was immersion, with an average score of 2.5. Indicating that this particular area may require further refinement and improvement in future iterations of the system. Together, these findings provide valuable insight into both the strengths and weaknesses of A Planet before our Time, helping guide future development decisions.

6 Discussion

The results of the usability study indicate that comfort was the highest-rated category, achieving an average score of 4.33. This suggests that participants generally found the system physically comfortable to use, highlighting a key strength of the overall experience. Closely following this, one of the highest individual item scores was related to hand tracking accuracy, which received an average of 4.30. This indicates that users perceived the tracking system as responsive and technically reliable.

However, despite this strong individual score, the overall hand tracking category was negatively impacted by a lower rating of 2.83 for the question concerning how natural the interaction felt. This discrepancy suggests that while the system performs well in terms of technical accuracy, it lacks a sense of realism—likely due to the absence of tactile feedback. As a result, the interaction may feel less convincing, even if it functions correctly.

The second highest-rated category overall was efficiency, with an average score of 3.91. Within this category, participants reported low levels of frustration when using the system (average 4.0) and relatively quick system responses (average 3.83). These findings indicate that users were generally able to interact with the environment in a smooth and manageable way, although there is still room for optimization.

Due to the lower score within the hand tracking category, ease of use emerged as the third highest-rated category, with an average score of 3.50. Participant feedback suggests that some users experienced difficulty understanding how to interact with the system, particularly in the absence of guidance or instructional cues. This highlights a need for clearer onboarding or in-game instructions to support first-time users.

Although the overall results did not fully meet expectations, they demonstrate that hand tracking is a promising interaction method with significant potential for further development. It is also important to acknowledge that the relatively small sample size, as well as the specific design of the questionnaire, may have influenced the overall results. Future studies with a larger participant group and refined evaluation methods could provide a more comprehensive assessment of usability.

7 Conclusion

The study evaluated the usability score of a real-time hand tracking VR experience in different categories, resulting in a relatively average score but indicating that this certain mechanic offers comfort and efficiency. However the application lacks in realism and immersion. Noting that the group of participants is rather small, further improvements also call for a bigger group of participants.

References

Biswas and Visell, Haptic Perception, Mechanics, and Material Technologies for Virtual Reality. Adv. Funct. Mater.2021, 31, 2008186. https://doi.org/10.1002/adfm.202008186.

Buckingham G (2021) Hand Tracking for Immersive Virtual Reality: Opportunities and Challenges. Front. Virtual Real. 2:728461. doi: 10.3389/frvir.2021.728461.

Holderied, H (2017): Evaluation of Interaction Concepts in Virtual Reality Applications. INFORMATIK 2017. Gesellschaft für Informatik, Bonn. PISSN: 1617-5468. ISBN: 978-3-88579-669-5. pp. 2511-2523. Studierendenkonferenz Informatik 2017 (SKILL 2017). Chemnitz. 25.-29. September 2017. doi: 10.18420/in2017_254

Masurovsky, A., Chojecki, P., Runde, D., Lafci, M., Przewozny, D., & Gaebler, M. (2020). Controller-Free Hand Tracking for Grab-and-Place Tasks in Immersive Virtual Reality: Design Elements and Their Empirical Study. Multimodal Technologies and Interaction, 4(4), 91. https://doi.org/10.3390/mti4040091.

Sharp, Keskin, Robertson, Taylor, Shotton, Kim, Rhemann, Leichter, Vinnikov, Yi. Wei, Freedman, Kohli, Krupka, Fitzgibbon and Shahram Izadi. 2015. Accurate, Robust, and Flexible Real-time Hand Tracking. In Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15). Association for Computing Machinery, New York, NY, USA, 3633–3642. https://doi.org/10.1145/2702123.2702179.

Voigt-Antons, Kojic, Ali and Möller, "Influence of Hand Tracking as a Way of Interaction in Virtual Reality on User Experience," 2020 Twelfth International Conference on Quality of Multimedia Experience (QoMEX), Athlone, Ireland, 2020, pp. 1-4, doi: 10.1109/QoMEX48832.2020.9123085.